Used Car Pricing Model Using a Ridge Regression

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The used car data provided was used to train a Machine Learning model to predict the fair market price of car listings. The data provided had a significant number of missing values, seen in the table below.

Missing Values per Column:

Drivetrain	136
Fuel Tank Capacity	113
Engine	80
Max Torque	80
Max Power	80
Length	64
Width	64
Height	64
Seating Capacity	64

In order to alleviate the model of dealing with so much missing data, the top ten most important features were kept in the data set. These features are the most relevant when determining the value of the car, and include the following set of columns.

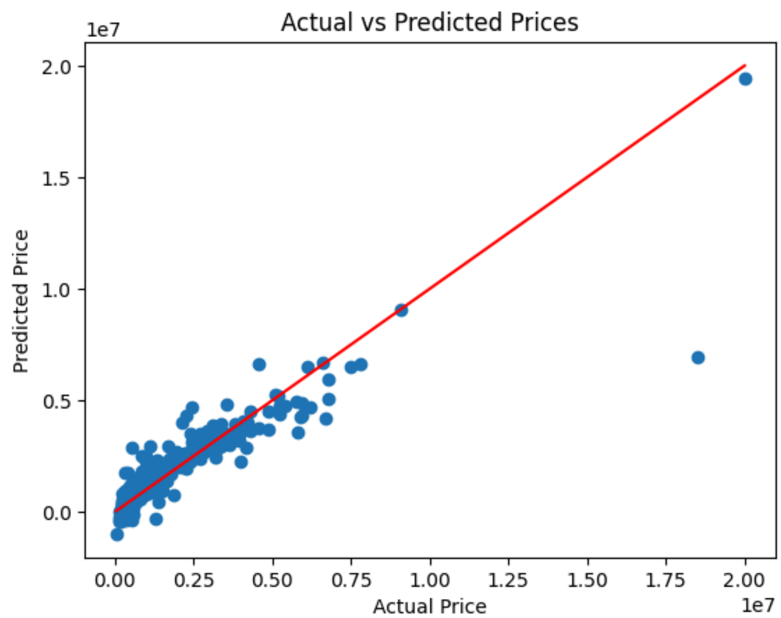
```
["Year", "Make", "Model", "Kilometer", "Engine", "Max Power",
"Transmission", "Fuel Type", "Owner", "Location"]
```

Other parameters were omitted for model simplicity, and to eliminate columns with an extreme amount of missing data. For example, the contributions of seating capacity, length, width, and height are also included with make and model. Max torque and max power are similar features, so only one is needed. The ten features listed above have the most significant impact on the listing price of a used car, so they are used as the predictors in this model. From this list, only two predictors have missing values, which are Engine and Max Power. These are the same 80 rows, making up less than 4% of the data, so they were deleted from the data set to move forward without any missing values.

To help with the model, the following features were made into integer or float type: Year, Kilometer, Engine, and Max Power. Setting the target to be “Price”, and the others as predictors, the model made the other categorical categories into numeric binary by the pd.get_dummies() function. This resulted in over 1000 unique categories, but was necessary to do before training the regression model. The data was split into 80/20 training and testing sets, and scaled by the StandardScaler from scikitlearn.

The Ridge Regression model was trained with the training data. For the Ridge Regression, the alpha parameter was optimized using Ridge_CV and found to be 67. Based on this model and the testing data, the following evaluation metrics were recorded.

```
MAE: 401629.91
RMSE: 809180.04
R²: 0.832
```



Most of the data is clustered, and follows a relatively linear trend. More complex models would likely overfit the data, and an R-Squared of 0.832 is okay for this model in predicting fair market prices. Any data that has a error margin that is far way from this predicted line should be analyzed by the group, but any data that is relatively close to this prediction would be fine to leave unchecked.